

Technical Datasheet of WIN-4G-GW-16UI8UO-ETH Module (Modbus TCP/IP)

1. Overview

The Universal Input (UI) and Universal Output (UO) board support **communication exclusively over Modbus TCP/IP**, enabling seamless control and monitoring of inputs/outputs. The board provide **configurable input and output selection via TCP/IP web pages**, ensuring flexibility across various applications. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.

Digital IO module is sturdy, low power usage and easy to use.

Through the **4G network, device data can be transmitted using the MQTT protocol**, allowing real-time monitoring and integration with cloud platforms or IoT systems.

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.



Specifications

<u>Category</u>	<u>UIUO Module</u>
Input Type	Analog Input (10V) Analog Input (5V) Digital Input NTC Sensor Input. (Configurable via TCP/IP web page)
Output Types	Analog Output (AO), Digital Output (DO). (configurable via TCP/IP web pages)
Connectors	Single port handling multiple input/output types

General –

Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 9 – 24 V DC or 24 V AC/DC Typical – 24V DC @ 700mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



UI (Universal Inputs) –

Digital Input channels	16
Sense Voltage	3.3 – 12 VDC (Depends on the selected type of Input)
Digital Sensing Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Analog Input channels	16
Input Signal	0 – 5VDC / 0 – 10VDC / NTC (Web page selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 15 VDC min
AI input impedance	120Ω

UO (Universal Outputs) –

Digital Output channels	8
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55°C~+150°C

Analog Output channels	8
Analog Resolution	10 bits (0 to 10VDC)
Accuracy	± 2% Full scale
Isolation	Optically Isolated
Conversion Time	15 msec

Additional Features: -

Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air

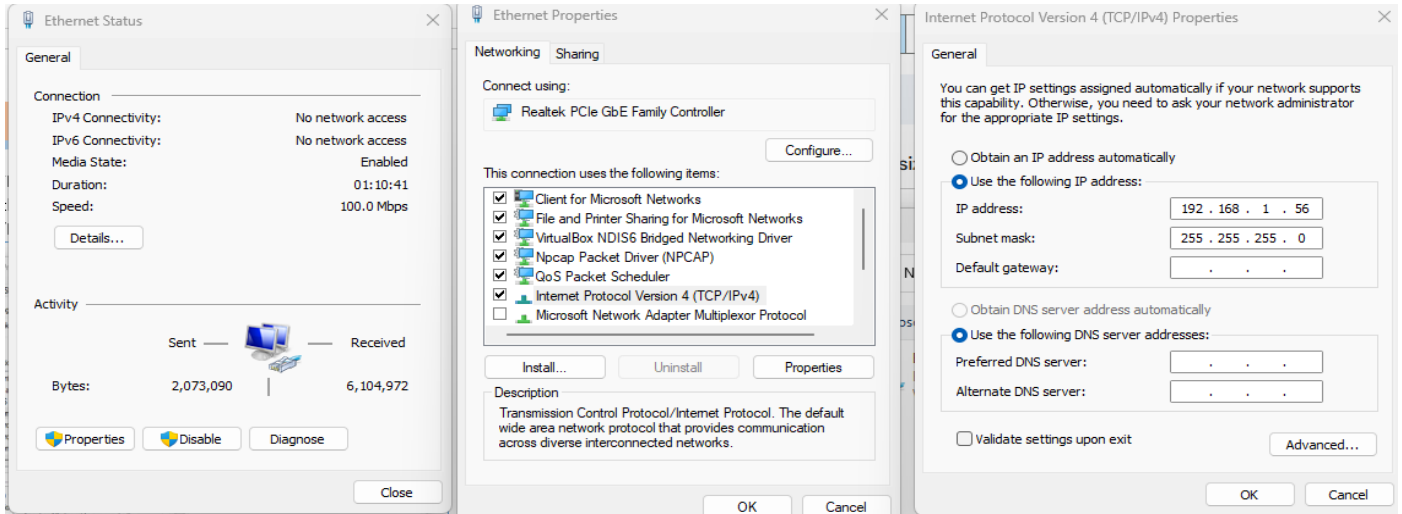
Configuration Of UIUO module

The UI/UO module supports flexible I/O configuration. Each of the 16 UI (Universal Input) ports can be independently configured as Analog Input, NTC Sensor Input, Digital Input. Similarly, each of the 8 UO (Universal Output) ports can be independently configured as, Analog Output, Digital Output

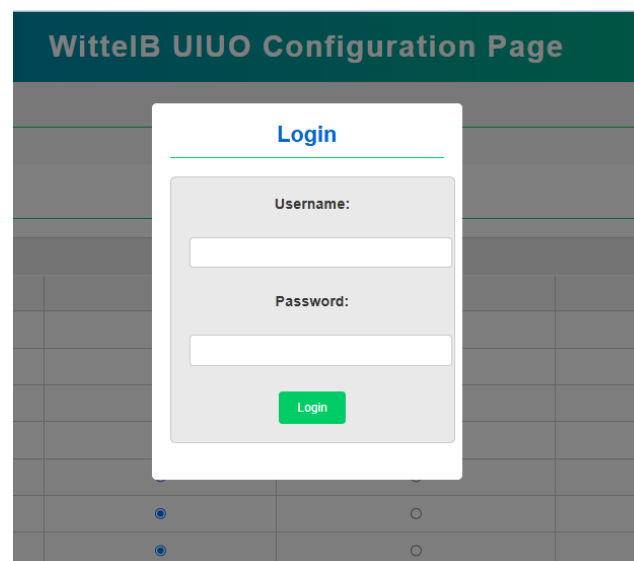
All the configuration can be done through Web page hosted at the default IP address (i.e. 192.168.1.177) or at the changed IP address.

The following are step by step guide for configuration:

1. Connect the Module to +24VDC POWER SUPPLY.
2. Connect the Board to the PC via an Ethernet Cable.
3. Use the default setting for PC as TCP/IPv4 (i.e., Address -192.168.1.56).



4. Now, open web browser and type **192.168.1.177** (default IP address) and Press 'Enter'. A login page will appear as shown below.



5. Default Login Credentials are :
Username: admin
Password: admin

6. After successful login, The UIUO Configuration page will be visible. It has multiple tables. The first named as 'Update UI Status' and 'Update UO Status' which have selectable radio button for doing the configuration of respective channels as shown below.

Update UI Status

CHANNEL	DI	10V	NTC	5V
1	☐	☒	☐	☐
2	☐	☐	☒	☐
3	☒	☐	☐	☐
4	☒	☐	☐	☐
5	☒	☐	☐	☐
6	☒	☐	☐	☐
7	☒	☐	☐	☐
8	☒	☐	☐	☐
9	☒	☐	☐	☐
10	☒	☐	☐	☐
11	☒	☐	☐	☐
12	☒	☐	☐	☐
13	☒	☐	☐	☐
14	☒	☐	☐	☐
15	☒	☐	☐	☐
16	☒	☐	☐	☐

Update UI Status

Update UO Status

CHANNEL	DO	AO
1	☐	☒
2	☐	☒
3	☐	☒
4	☐	☒
5	☐	☒
6	☐	☒
7	☐	☒
8	☐	☒

Update UO Status

7. For example, to set Channel 1 of UI as NTC Sensor input, the radio button in front of Channel 1 must be selected as NTC and then "Update UI Status" button need to be pressed. This will make the Channel 1 of UI port as NTC sensor input port.

Note: Always restart the device/refresh the page after doing configuration for UI/UO Status, then only changes in configuration will take place.

8. The third table “UI_DATA” is for viewing the UI port data locally on the web page.

Data Section	
UI_DATA	
CHANNEL	UI_DATA
1	421
2	117
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0

9. User can select all the desired configuration at once and then press the Update button for UI or UO to update the Port status in the module.
10. User can change the IP and Mac Address for the TCP from the tab “MODBUS TCP CONFIGURATION” in the web page and press the “Save Configuration” button and restart the device to set the new IP, as shown below.

UIUO Configuration Page

UIUO PAGE MODBUS TCP CONFIGURATION MQTT CONFIGURATION RTU CONFIGURATION SLAVE CONFIGURATION UPDATE LOGIN

Modbus TCP Configuration

MAC ID:
 DE AD BE EF FE ED

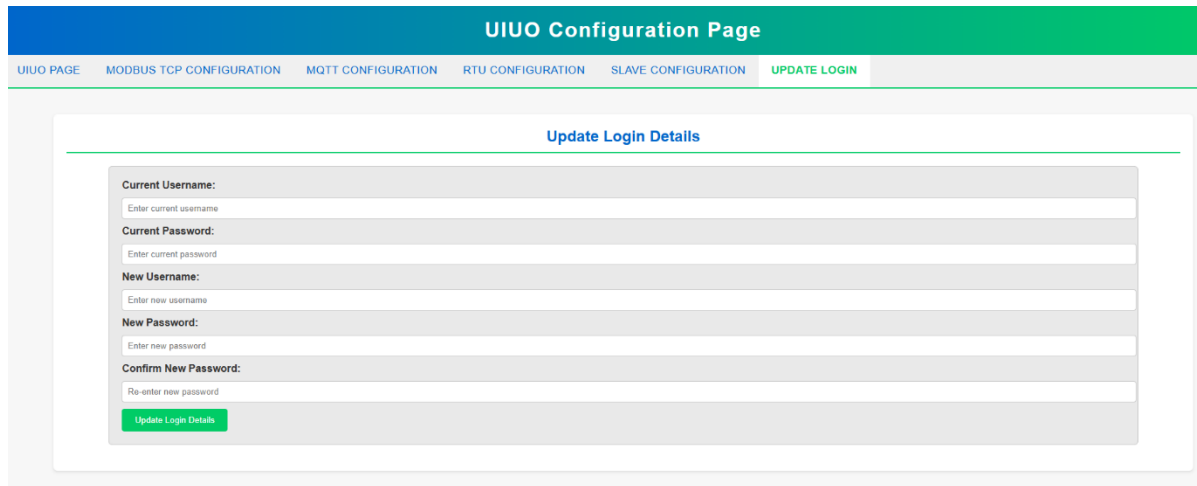
IP Address:
 192.168.1.177

Port:
 502

Save Configuration

Note: To reset the IP address to default IP, the user need to press and hold the reset button located near the Ethernet port for 5 seconds.

11. On the tab “UPDATE LOGIN” page, the user can change the Username and Password, to enhance the security of UIUO module, as show below.



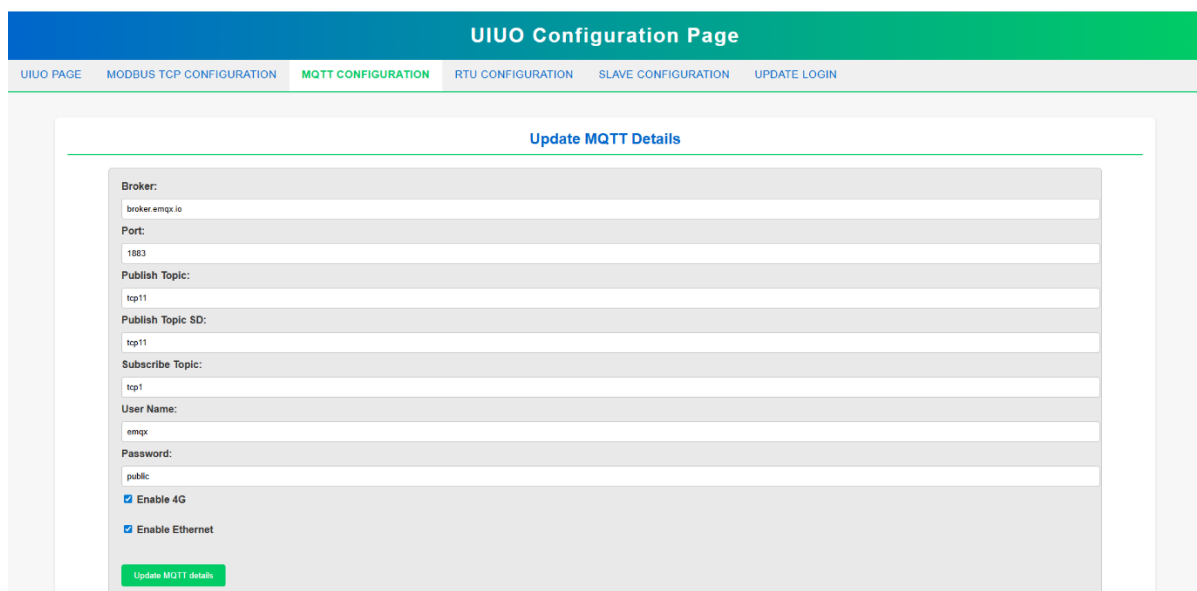
The screenshot shows the 'UIUO Configuration Page' with the 'UPDATE LOGIN' tab selected. The 'Update Login Details' form contains the following fields:

- Current Username:** Enter current username
- Current Password:** Enter current password
- New Username:** Enter new username
- New Password:** Enter new password
- Confirm New Password:** Re-enter new password

A green 'Update Login Details' button is located at the bottom of the form.

12. After all the configuration, User can now connect the UIUO module to Modbus TCP master at the set IP address or default IP address (192.168.1.177).

13. On the tab “MQTT CONFIGURATION” page, the user can give the MQTT Credentials to get the data on cloud, and select the option to get data on cloud through 4G/Ethernet, then click update MQTT details.

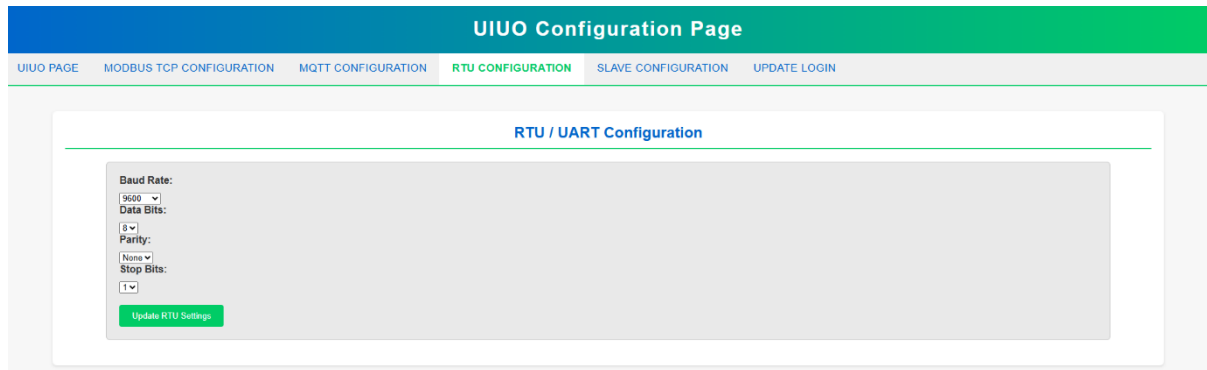


The screenshot shows the 'UIUO Configuration Page' with the 'MQTT CONFIGURATION' tab selected. The 'Update MQTT Details' form contains the following fields:

- Broker:** broker.emqx.io
- Port:** 1883
- Publish Topic:** tcp11
- Publish Topic SD:** tcp11
- Subscribe Topic:** tcp1
- User Name:** emqx
- Password:** public
- ☒ Enable 4G
- ☒ Enable Ethernet

A green 'Update MQTT details' button is located at the bottom of the form.

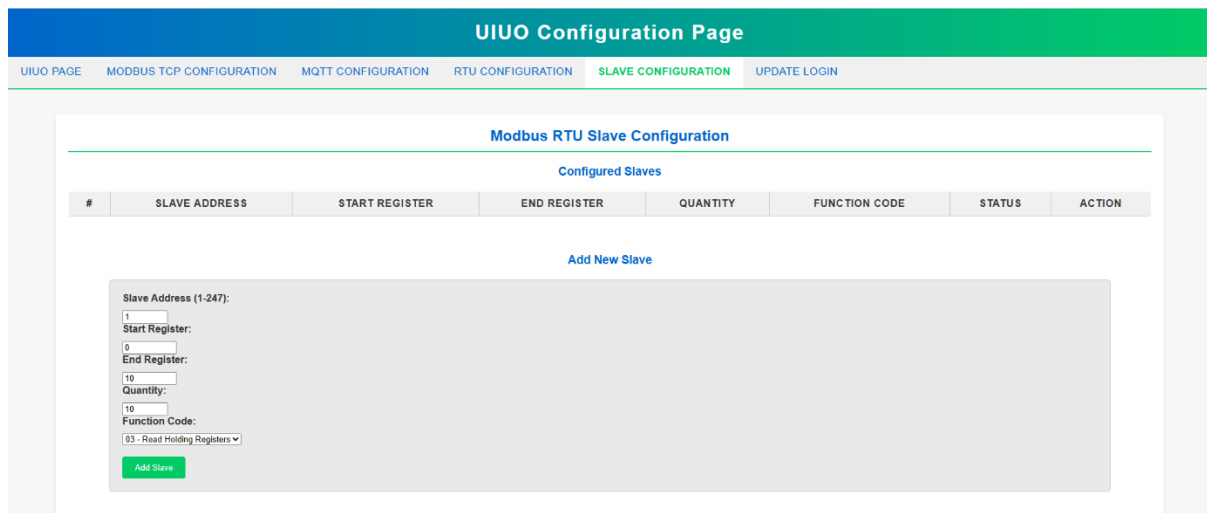
14. On the tab “RTU CONFIGURATION” page, user can give the RTU Configuration for slave - Baudrate, Data bit, Parity, Stop bit, then submit the configuration.



The screenshot shows the "UIUO Configuration Page" with a green header. The navigation bar includes: UIUO PAGE, MODBUS TCP CONFIGURATION, MQTT CONFIGURATION, **RTU CONFIGURATION**, SLAVE CONFIGURATION, and UPDATE LOGIN. The main content area is titled "RTU / UART Configuration". It contains a form with the following fields:

- Baud Rate: 9600 (dropdown)
- Data Bits: 8 (dropdown)
- Parity: None (dropdown)
- Stop Bits: 1 (dropdown)
- Update RTU Settings (button)

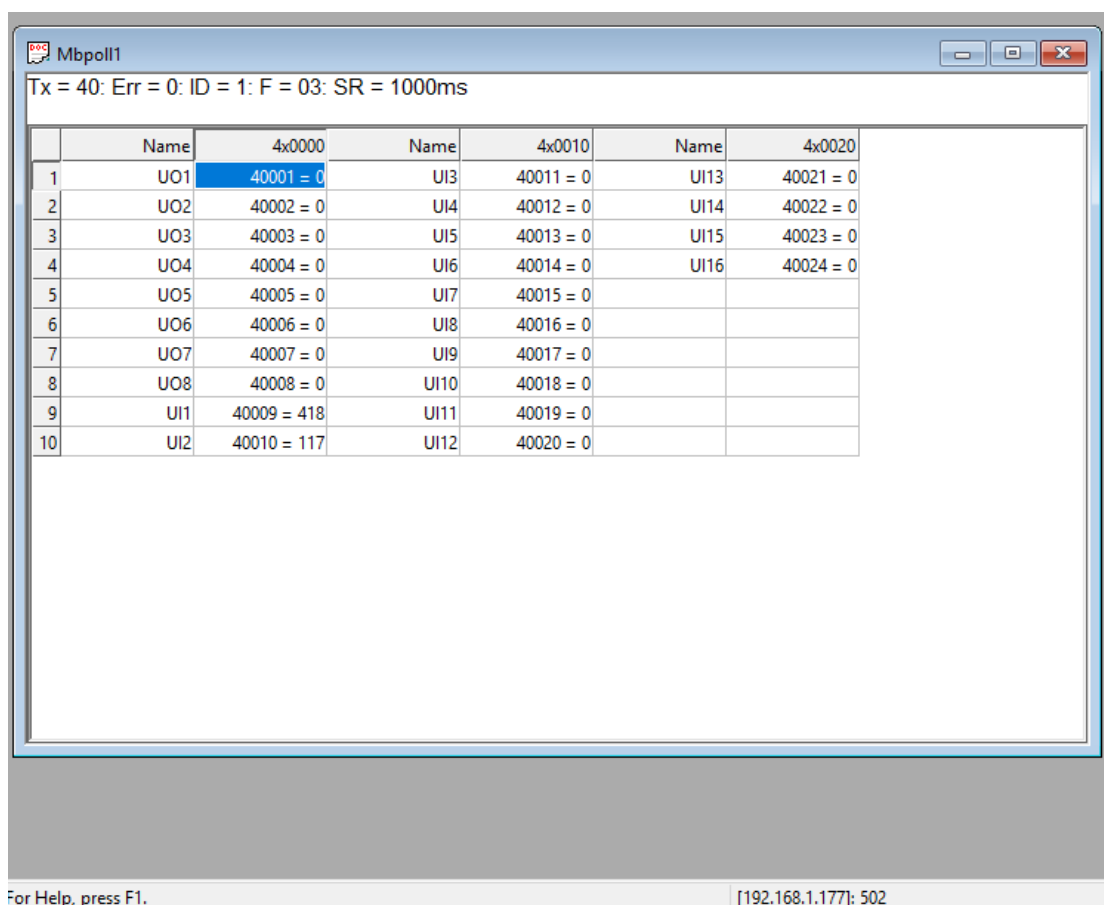
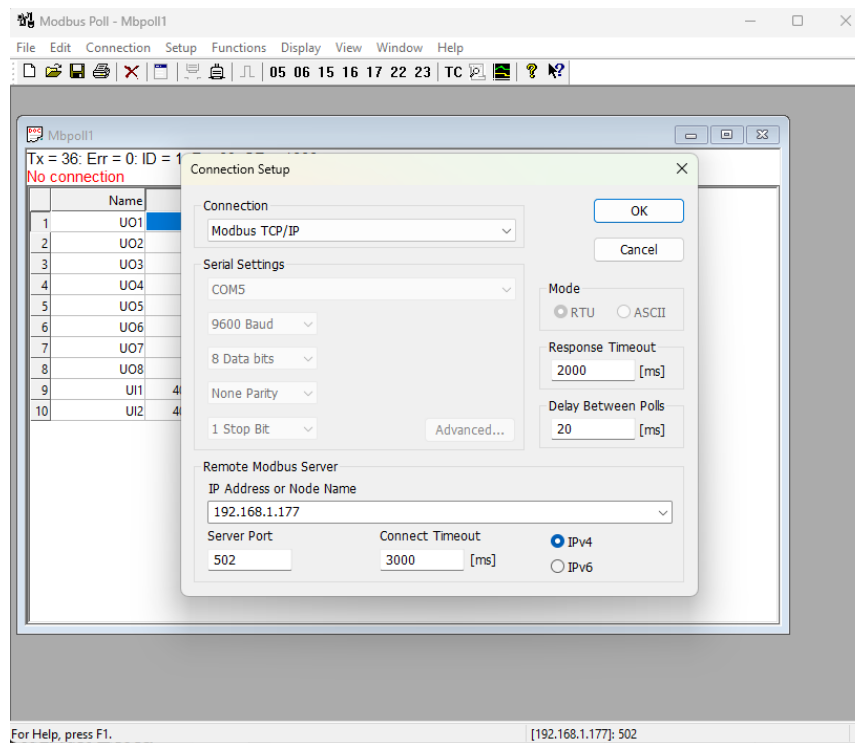
15. On the tab “SLAVE CONFIGURATION” page, user can give the Slave Configuration as slave ID, Start Register, End Register, Quantity, Function Code, then submit the configuration.



The screenshot shows the "UIUO Configuration Page" with a green header. The navigation bar includes: UIUO PAGE, MODBUS TCP CONFIGURATION, MQTT CONFIGURATION, RTU CONFIGURATION, **SLAVE CONFIGURATION**, and UPDATE LOGIN. The main content area is titled "Modbus RTU Slave Configuration". It contains a table titled "Configured Slaves" with the following columns: #, SLAVE ADDRESS, START REGISTER, END REGISTER, QUANTITY, FUNCTION CODE, STATUS, and ACTION. Below the table is a section titled "Add New Slave" with a form containing the following fields:

- Slave Address (1-247): 1
- Start Register: 0
- End Register: 10
- Quantity: 10
- Function Code: 03 - Read Holding Registers (dropdown)
- Add Slave (button)

16. Here we have shown a connecting with Modbus Poll software for demonstration in the below images.



17. The data of UI port can be fetched from the mentioned Modbus register in the table below, irrespective of DI, AI, NTC, selection. The Desired Inputs are as follows:

INPUT TYPE	Input Value
DI	0 V(0) , 12VDC (1)
AI(5V)	0 to 5VDC (0-1023)
AI(10V)	0 to 10VDC (0-1023)
NTC	NTC sensor(Temperature in °C)

18. Similarly, the data of the UO port can be set in the same register as mentioned in the table below. For DO, It should be sent 0 or 1 (12VDC). And for AO a 10-bit resolution value must be sent (i.e. from 0 to 1023) to set output of 0 to 10VDC.

OUTPUT Type	Output Value
DO	0 (0V) , 1 (12VDC)
AO	0 to 10VDC (0-1023)

MODBUS COILS TABLE FOR UIUO PORTS:-

ID	Function Description	Register Description	Modbus Function Code	Data Type
1	UO 1 (DO)	00001	0x05,0x15,0x01	Boolean
2	UO 2 (DO)	00002	0x05,0x15,0x01	Boolean
3	UO 3 (DO)	00003	0x05,0x15,0x01	Boolean
4	UO 4 (DO)	00004	0x05,0x15,0x01	Boolean
5	UO 5 (DO)	00005	0x05,0x15,0x01	Boolean
6	UO 6 (DO)	00006	0x05,0x15,0x01	Boolean
7	UO 7 (DO)	00007	0x05,0x15,0x01	Boolean
8	UO 8 (DO)	00008	0x05,0x15,0x01	Boolean
9	UI 1 (DI)	00009	0x01	Boolean
10	UI 2 (DI)	00010	0x01	Boolean
11	UI 3 (DI)	00011	0x01	Boolean
12	UI 4 (DI)	00012	0x01	Boolean

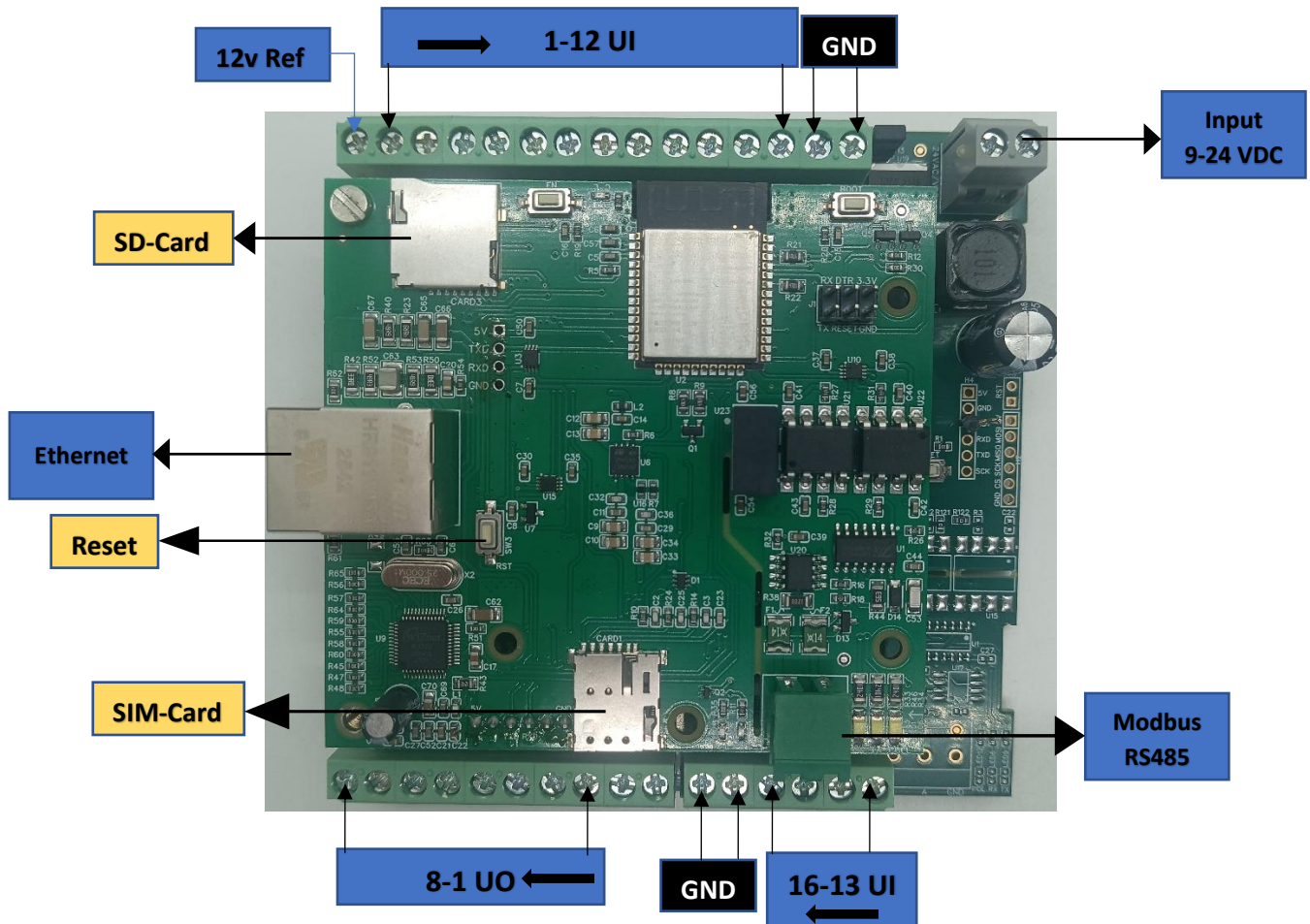
13	UI 5 (DI)	00013	0x01	Boolean
14	UI 6 (DI)	00014	0x01	Boolean
15	UI 7 (DI)	00015	0x01	Boolean
16	UI 8 (DI)	00016	0x01	Boolean
17	UI 9 (DI)	00017	0x01	Boolean
18	UI 10 (DI)	00018	0x01	Boolean
19	UI 11 (DI)	00019	0x01	Boolean
20	UI 12 (DI)	00020	0x01	Boolean
21	UI 13 (DI)	00021	0x01	Boolean
22	UI 14 (DI)	00022	0x01	Boolean
23	UI 15 (DI)	00023	0x01	Boolean
24	UI 16 (DI)	00024	0x01	Boolean

MODBUS REGISTERS TABLE FOR UIUO PORTS:-

1	UO 1 (AO)	40001	0X03	16 Bit Unsigned int
2	UO 2 (AO)	40002	0X03	16 Bit Unsigned int
3	UO 3 (AO)	40003	0X03	16 Bit Unsigned int
4	UO 4 (AO)	40004	0X03	16 Bit Unsigned int
5	UO 5 (AO)	40005	0X03	16 Bit Unsigned int
6	UO 6 (AO)	40006	0X03	16 Bit Unsigned int
7	UO 7 (AO)	40007	0X03	16 Bit Unsigned int
9	UI 1 (AI,NTC)	40009	0x03	16 Bit Unsigned int
10	UI 2 (AI,NTC)	40010	0x03	16 Bit Unsigned int
11	UI 3 (AI,NTC)	40011	0x03	16 Bit Unsigned int
12	UI 4 (AI,NTC)	40012	0x03	16 Bit Unsigned int
13	UI 5 (AI,NTC)	40013	0X03	16 Bit Unsigned int
14	UI 6 (AI,NTC)	40014	0X03	16 Bit Unsigned int
15	UI 7 (AI,NTC)	40015	0X03	16 Bit Unsigned int
16	UI 8 (AI,NTC)	40016	0X03	16 Bit Unsigned int

Wiring Diagram:-

The Port number and Sequence of UI and UO are as mentioned in the wiring diagrams shown below. Also the position of default reset button is as shown in the diagram.



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